

Lecture 12 — Dementia

Companion Figures

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Figure L12.7

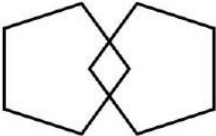
One point for each answer					DATE:			
ORIENTATION								
Year	Season	Month	Date	Time	/ 5	/ 5	/ 5	
Country	Town	District	Hospital	Ward/Floor	/ 5	/ 5	/ 5	
REGISTRATION								
Examiner names three objects (e.g. apple, table, penny) and asks the patient to repeat (1 point for each correct. THEN the patient learns the 3 names repeating until correct).					/ 3	/ 3	/ 3	
ATTENTION AND CALCULATION								
Subtract 7 from 100, then repeat from result. Continue five times: 100, 93, 86, 79, 65. (Alternative: spell "WORLD" backwards: DLROW).					/ 5	/ 5	/ 5	
RECALL								
Ask for the names of the three objects learned earlier.					/ 3	/ 3	/ 3	
LANGUAGE								
Name two objects (e.g. pen, watch).					/ 2	/ 2	/ 2	
Repeat "No ifs, ands, or buts".					/ 1	/ 1	/ 1	
Give a three-stage command. Score 1 for each stage. (e.g. "Place index finger of right hand on your nose and then on your left ear").					/ 3	/ 3	/ 3	
Ask the patient to read and obey a written command on a piece of paper. The written instruction is: "Close your eyes".					/ 1	/ 1	/ 1	
Ask the patient to write a sentence. Score 1 if it is sensible and has a subject and a verb.					/ 1	/ 1	/ 1	
COPYING: Ask the patient to copy a pair of intersecting pentagons								
					/ 1	/ 1	/ 1	
MMSE scoring								
24-30: no cognitive impairment								
18-23: mild cognitive impairment								
TOTAL:					/ 30	/ 30	/ 30	

Figure L12.7 — MMSE scoring form and thresholds

Figure L12.8

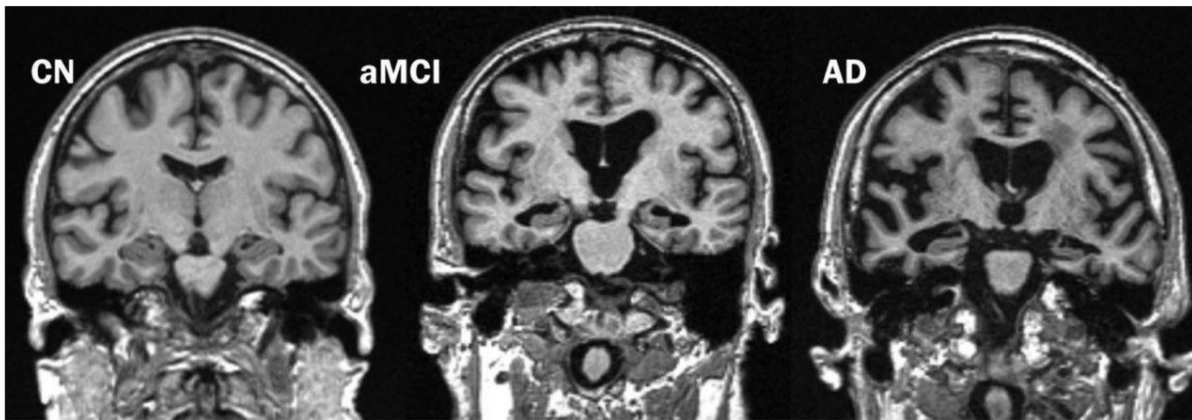
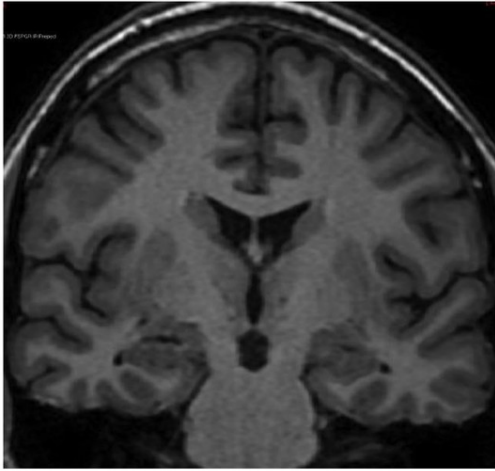
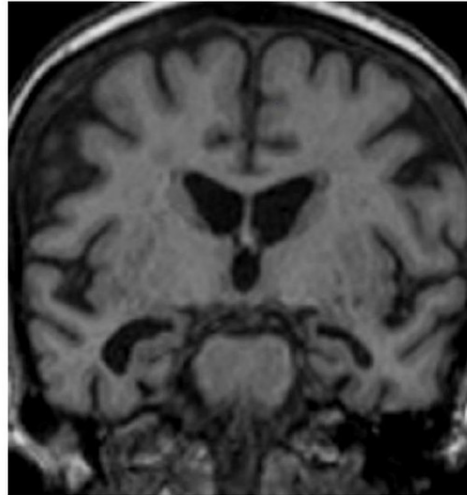


Figure L12.8 — sMRI progression: cognitively normal → aMCI → AD

Figure L12.14



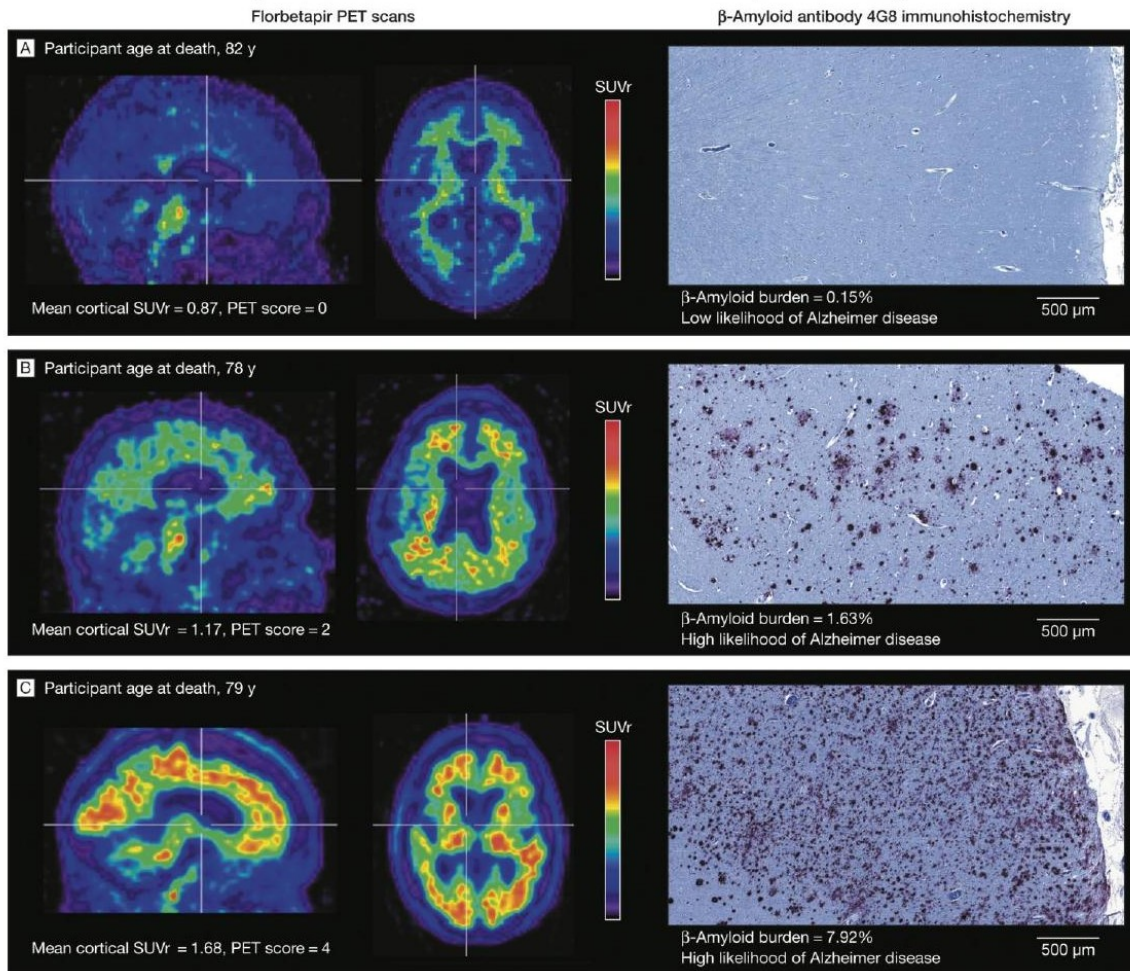
Healthy Elderly



Patient

Figure L12.14 — Healthy elderly vs AD patient (hippocampal atrophy visible)

Figure L12.9



Clark et al. Jama 2011

Figure L12.9 — ^{18}F -Florbetapir PET vs amyloid post-mortem (Clark 2011)

Figure L12.18

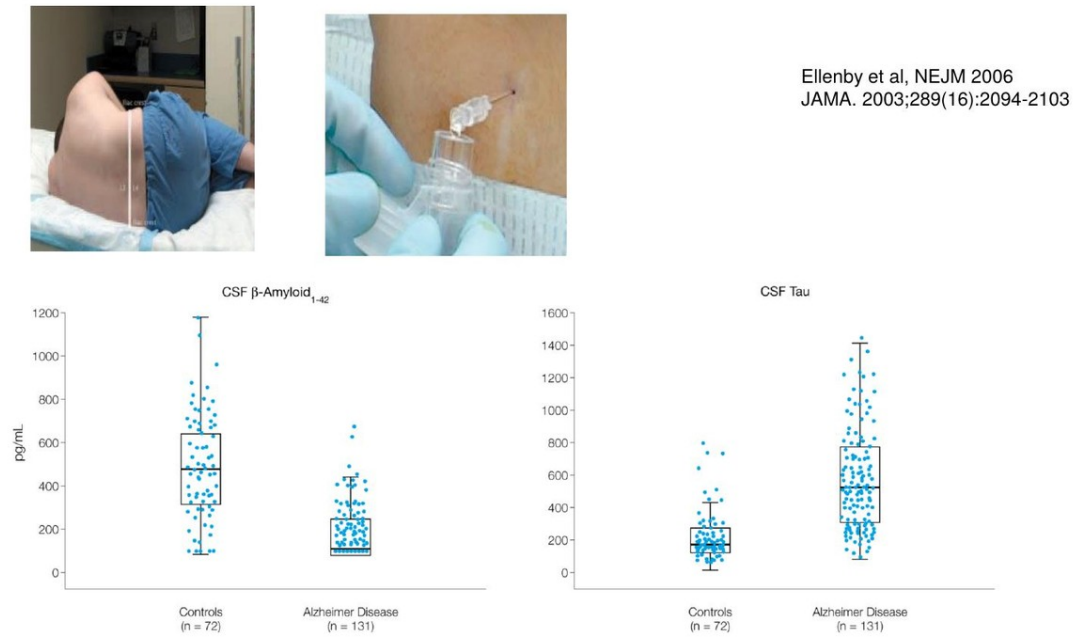


Figure L12.18 — CSF biomarkers in AD: A β low, tau high

Figure L12.16

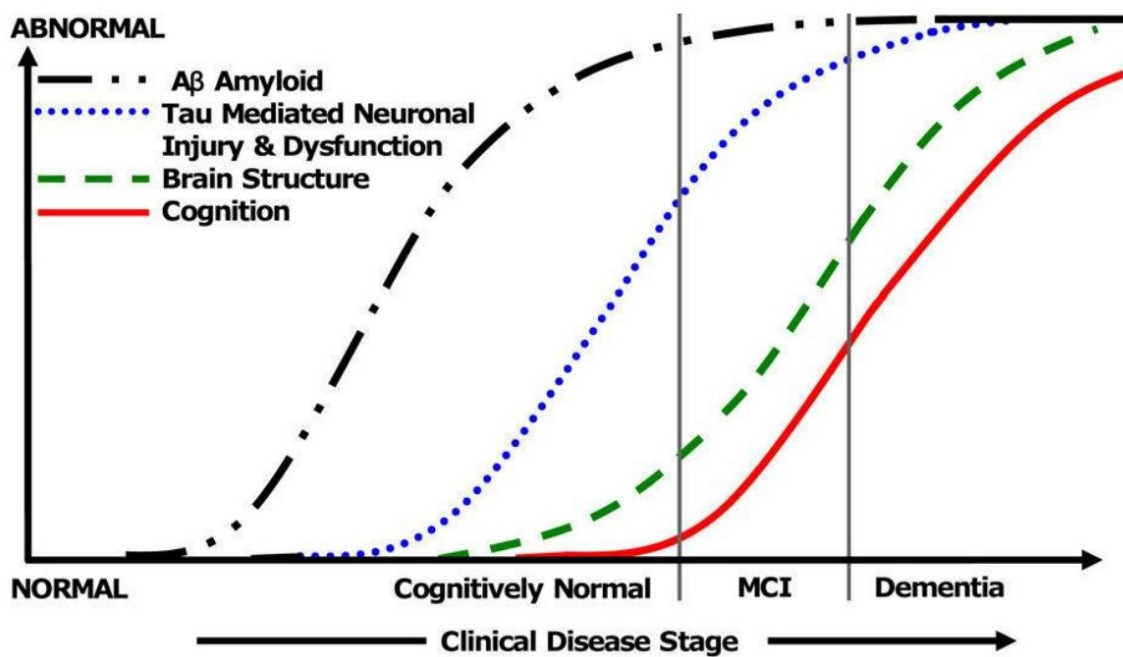


Figure L12.16 — Biomarker sequence in AD (Jack 2010): amyloid first, cognition last

Figure L12.6

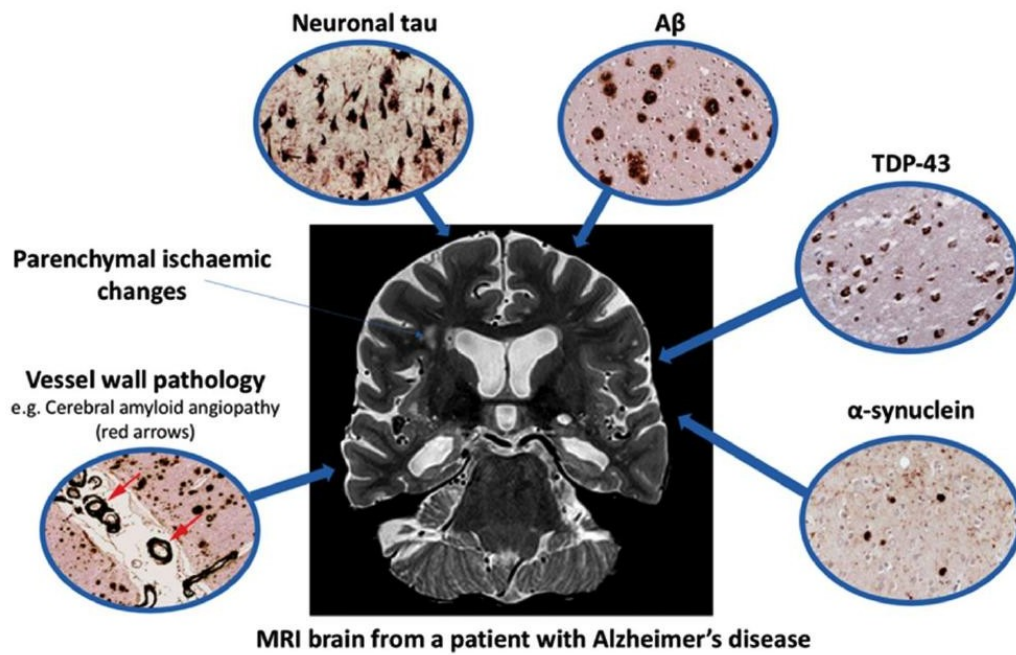


Figure L12.6 — Mixed pathology in an Alzheimer brain on MRI

Figure L12.10

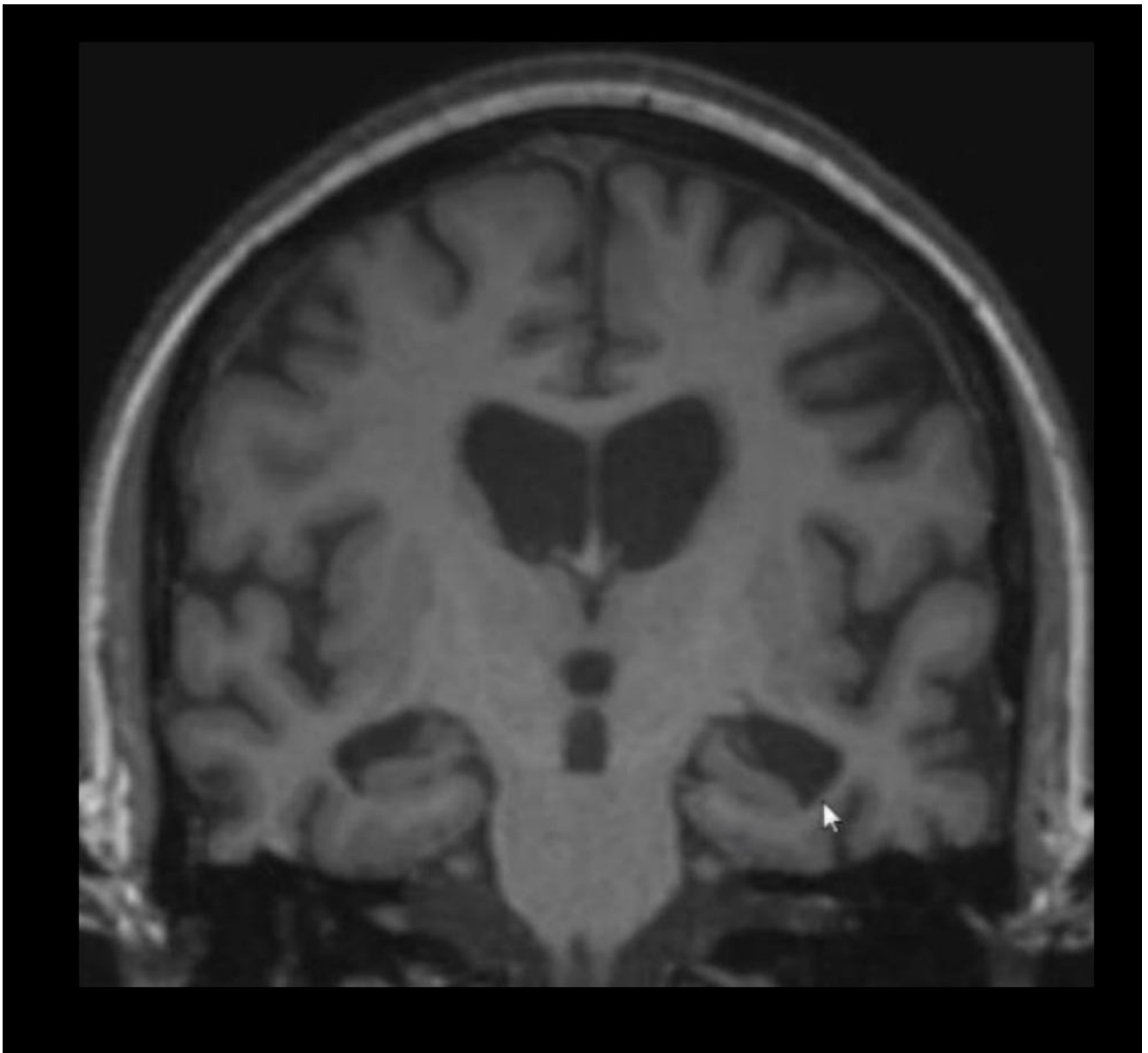


Figure L12.10 — AD patient coronal T1 MRI (bilateral hippocampal atrophy)

Figure L12.15

and tau (B) according to Thal and Braak

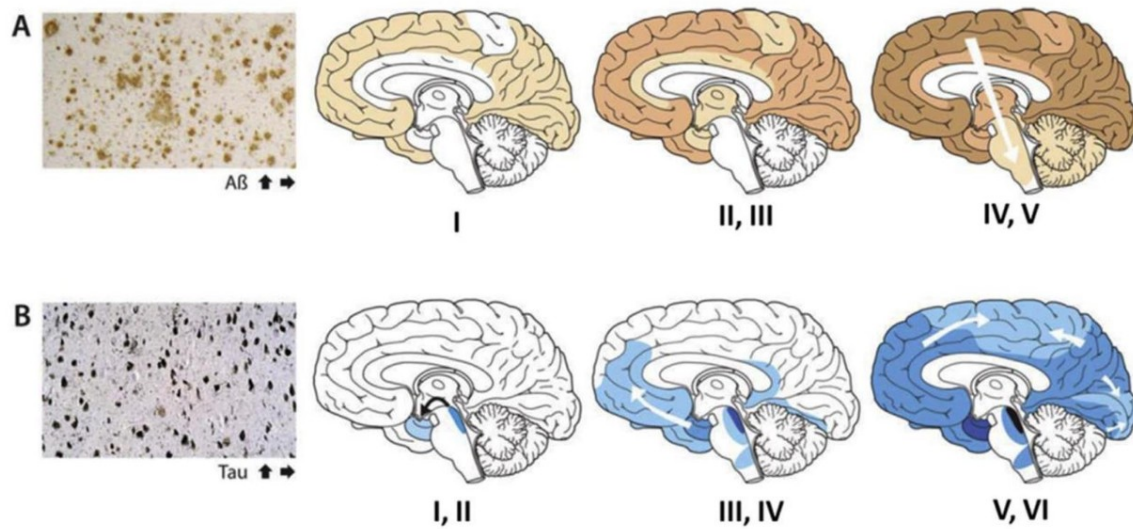


Figure L12.15 — Thal amyloid (I–V) and Braak tau (I–VI) staging

Figure L12.20

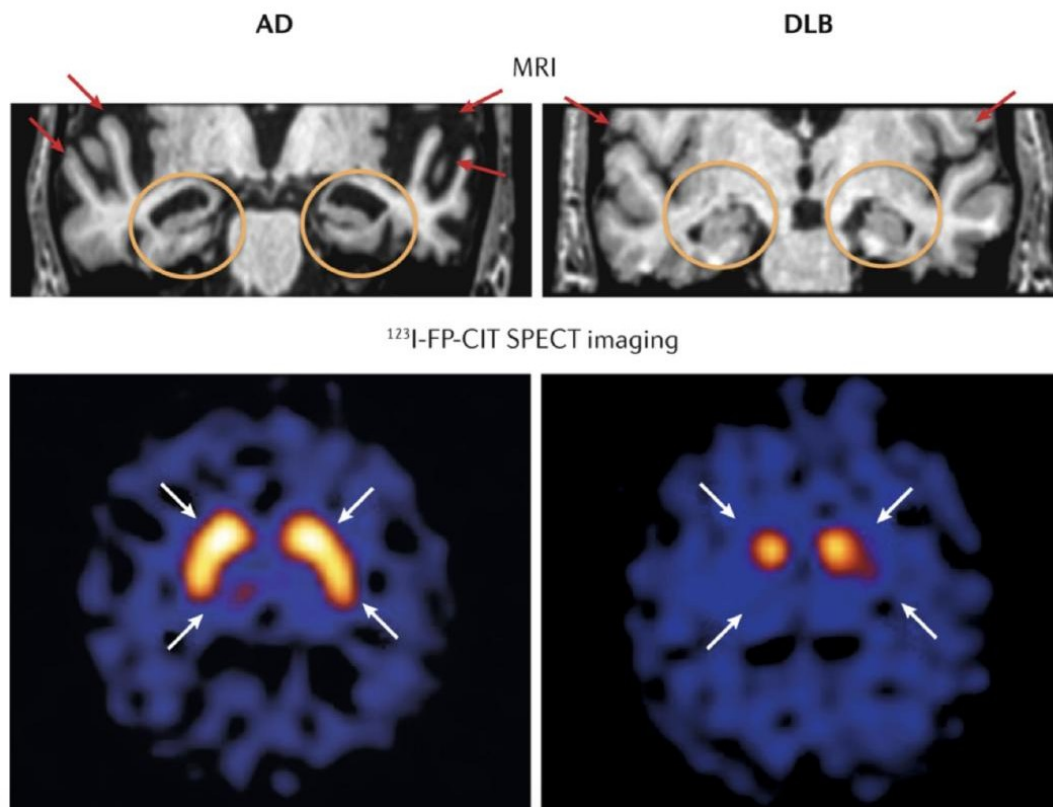


Figure L12.20 — AD vs DLB: MRI (hippocampal atrophy vs preserved) and ¹²³I-FP-CIT SPECT (comma vs dot)

Figure L12.21

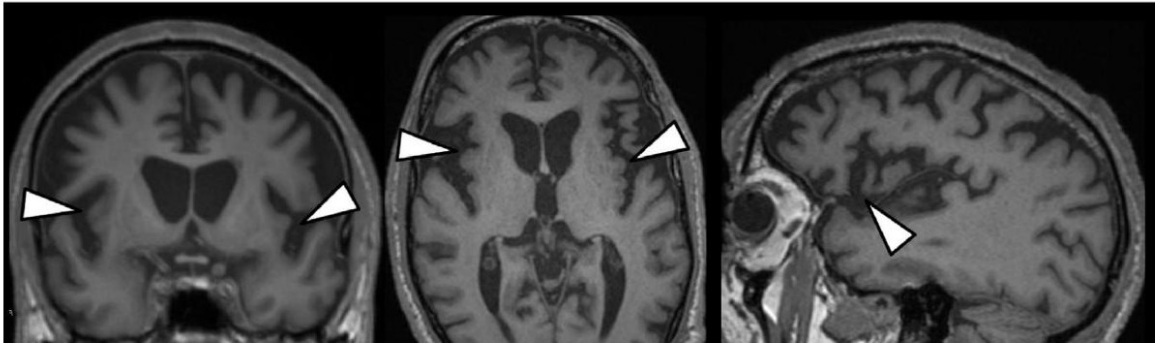


Figure L12.21 — FTD on MRI: bilateral perisylvian volume loss (C9orf72 case)

Figure L12.1

compared to late onset dementia

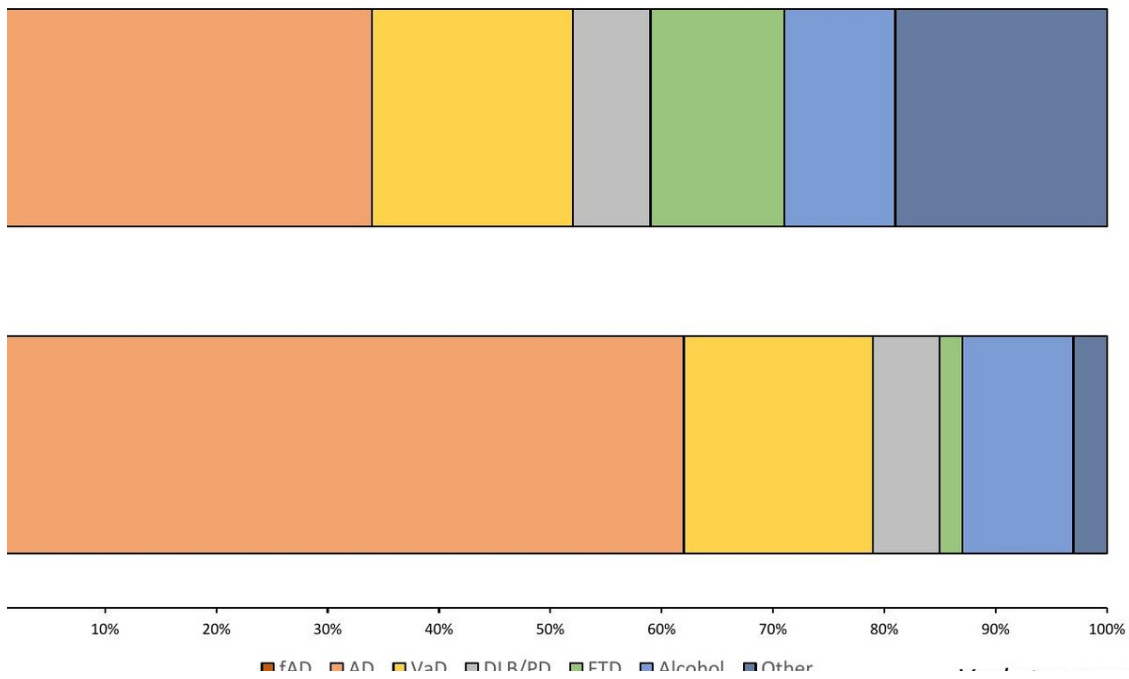


Figure L12.1 — YOD vs LOD aetiology proportions

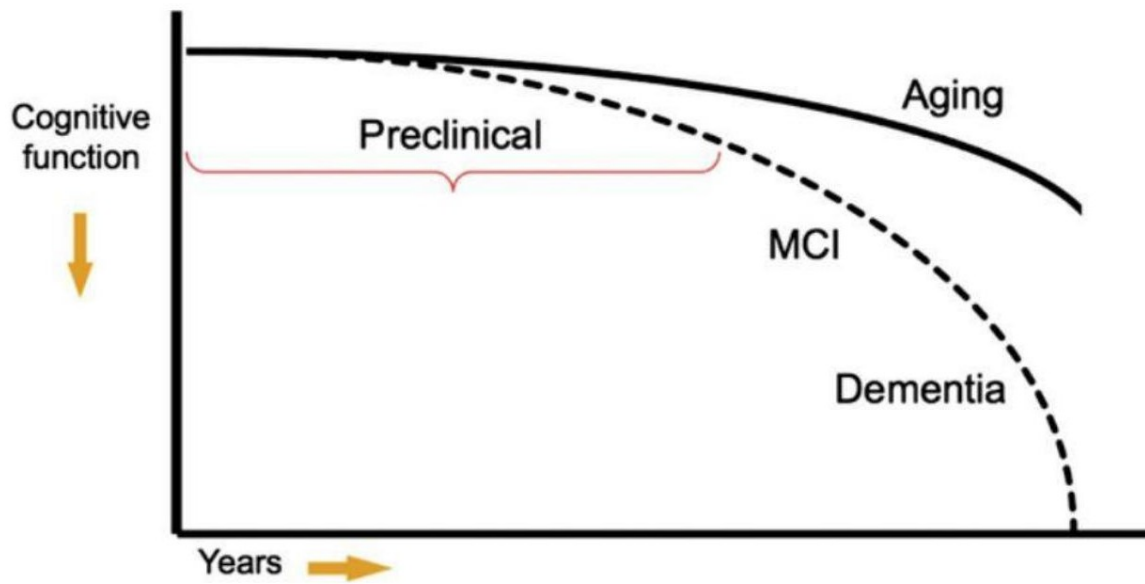
Figure L12.2

<i>Common</i>	<i>Potentially reversible</i>	<i>Rare</i>
Alzheimer's disease	Depression	Progressive supranuclear palsy
Vascular dementia	Alcohol related brain damage	Multiple system atrophy,
Frontotemporal dementia	Endocrine (hypothyroidism, Cushing's, Addison's)	Corticobasal degeneration
Dementia with Lewy bodies	B1/B12/B6 deficiency	Huntington's disease
	Benign tumours	Creutzfeldt–Jakob disease
	Normal pressure hydrocephalus	Progressive multifocal leucoencephalopathy
	Infections – HIV/syphilis, Whipple's disease	Subacute sclerosing panencephalitis
	Limbic encephalitis (paraneoplastic/autoimmune)	Leucodystrophies.
	Inflammatory (vasculitis, multiple sclerosis, sarcoid)	Neuronal ceroid lipofuscinosis (Kuf's disease).
		Cerebral autosomal dominant arteriopathy with subcortical infarcts and leucoencephalopathy (CADASIL).
		Mitochondrial encephalopathies

Figure L12.2 — Common, reversible and rare causes of dementia

Figure L12.3

THE CONTINUUM OF DEMENTIA



Sperling et al. *Alzheimer's & Dementia* 2011

Figure L12.3 — Continuum of dementia (Sperling 2011)

Figure L12.4

The realities of cognition in the clinic

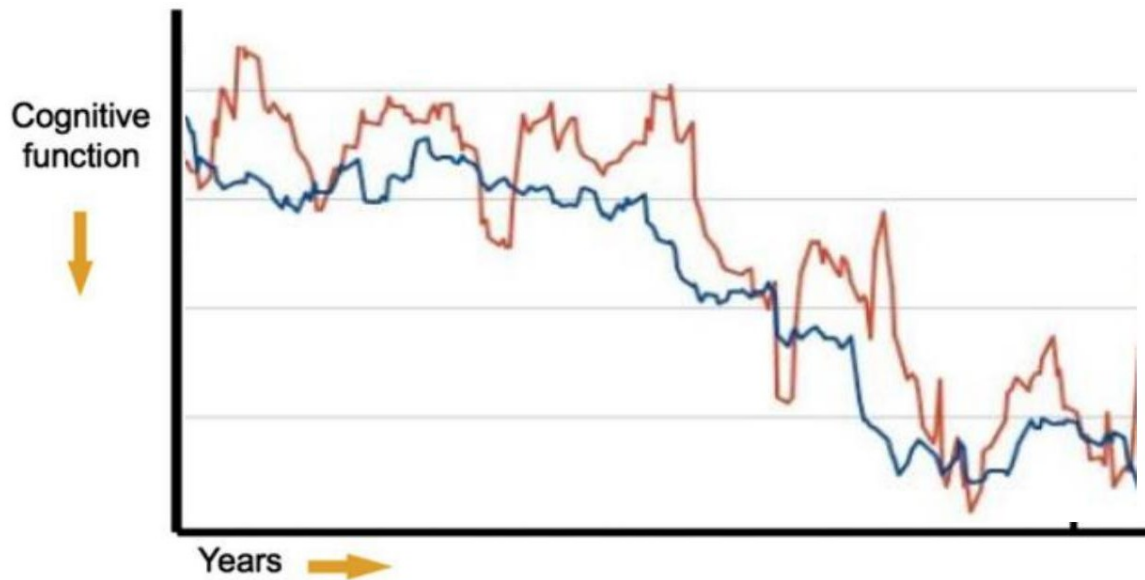
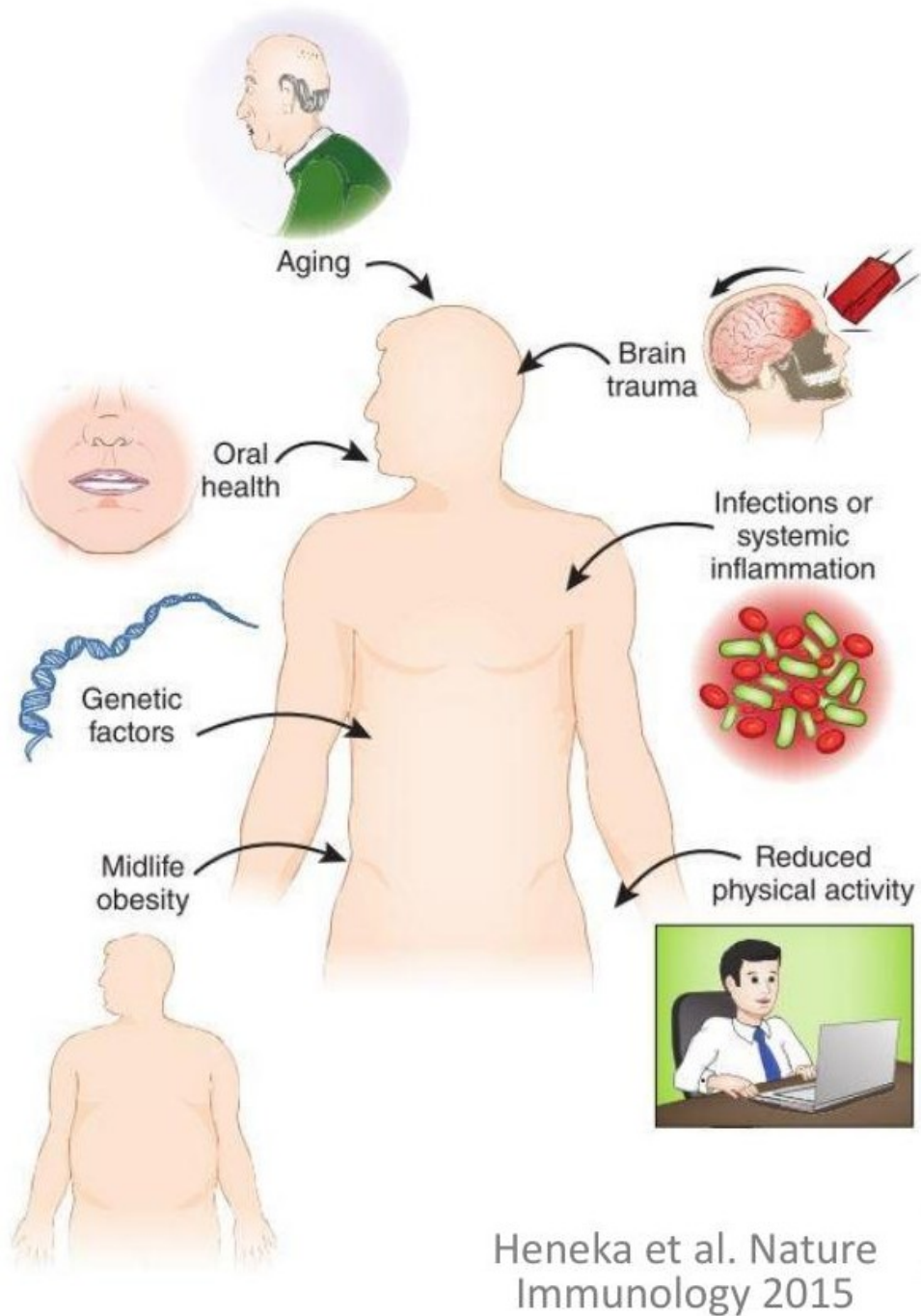


Figure L12.4 — Realities of cognition in the clinic

Figure L12.5



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Figure L12.5 — Risk factors for dementia (Heneka 2015)

Figure L12.11

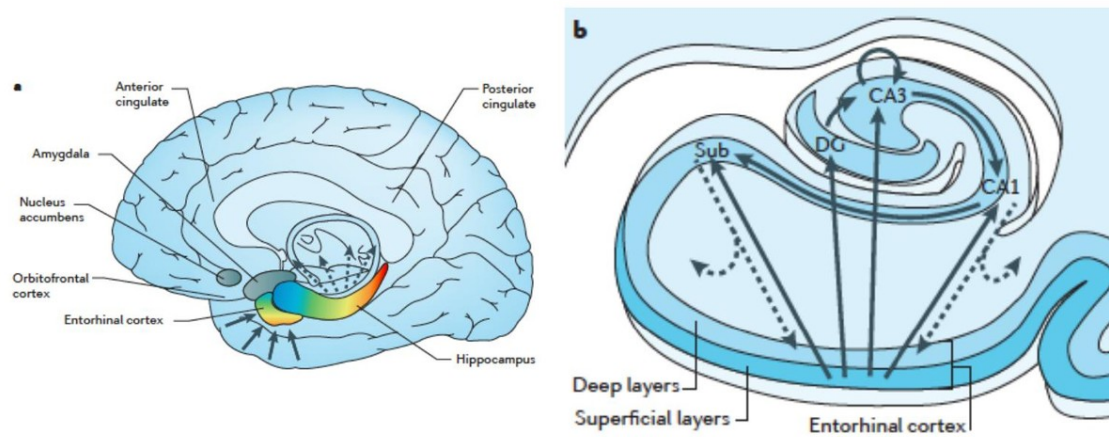
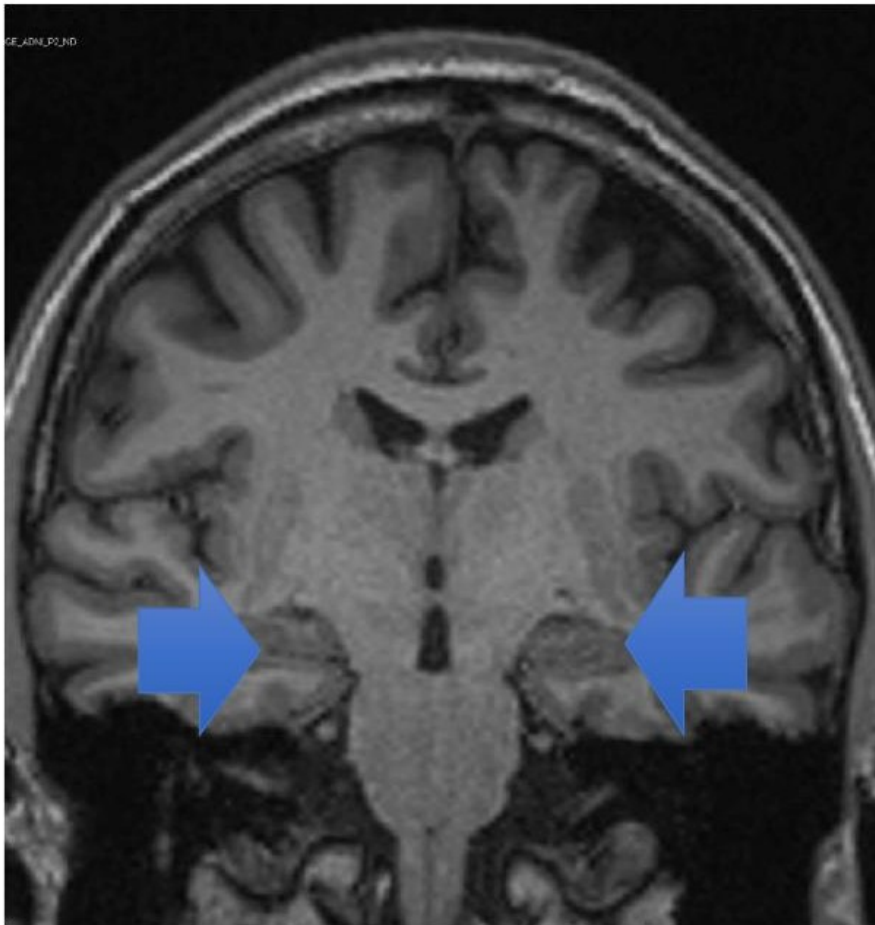


Figure L12.11 — Hippocampal circuitry (Small 2011)

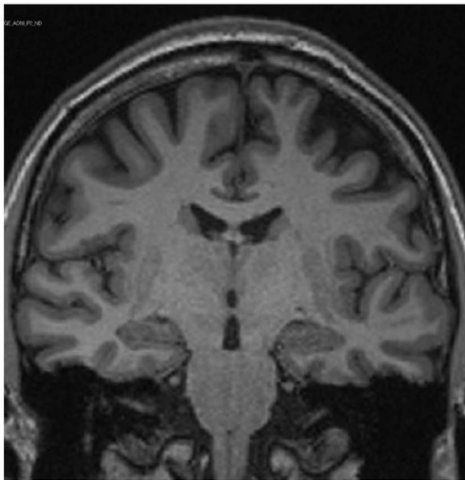
Figure L12.12



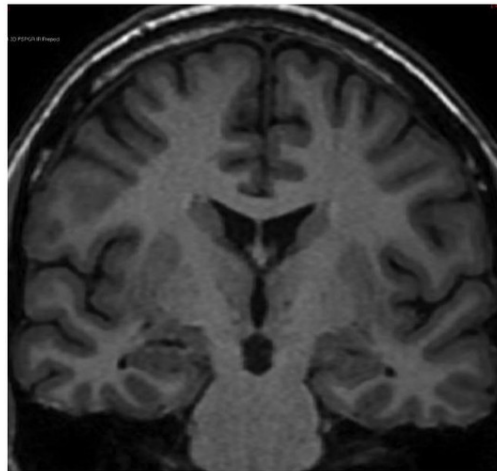
Normal Hippocampi (25 year old)

Figure L12.12 — Normal hippocampi (25yo reference)

Figure L12.13



25 year old



67 year old

Figure L12.13 — Age-related atrophy (25yo vs 67yo)

Figure L12.17

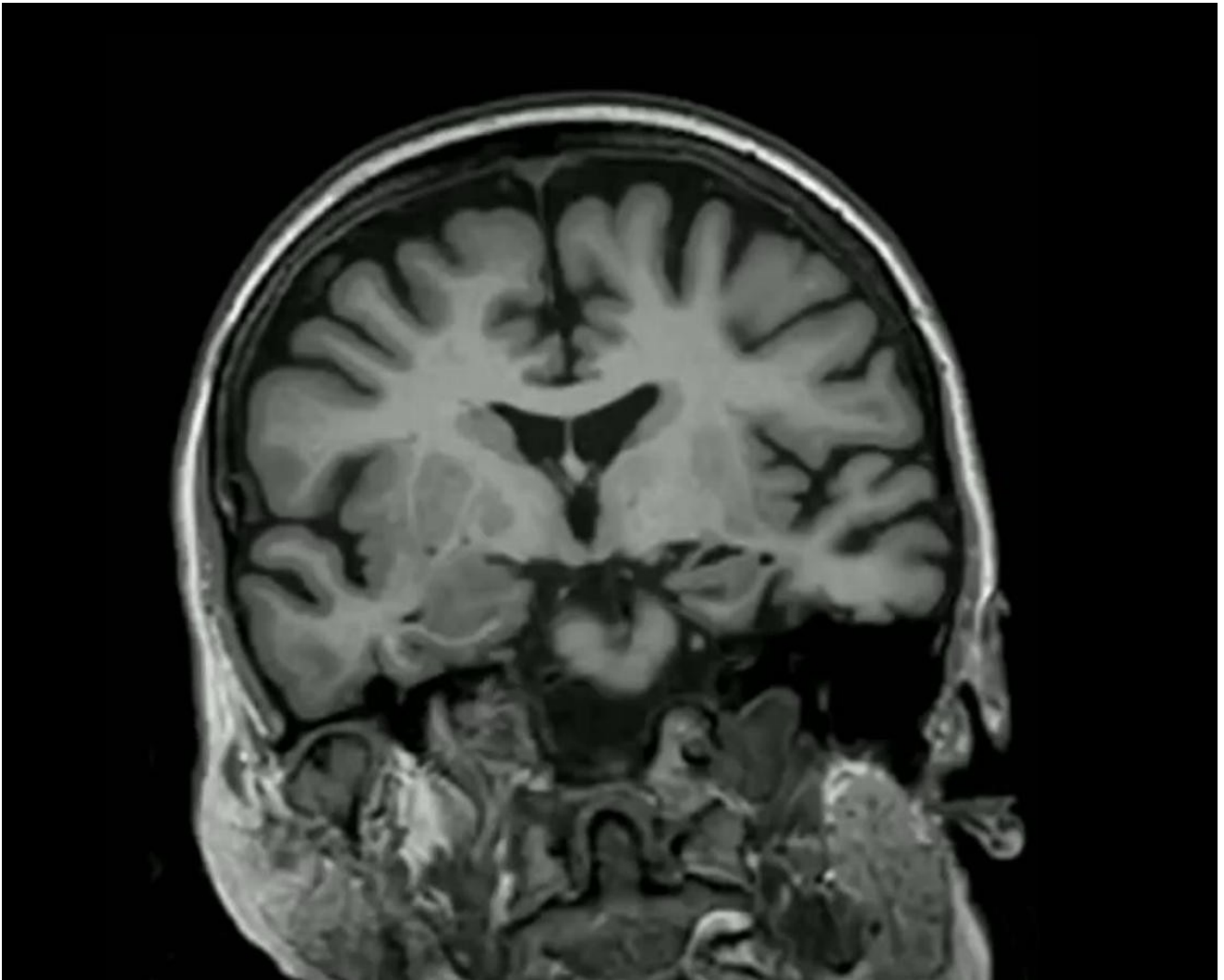


Figure L12.17 — Progressive AD atrophy (coronal T1)

Figure L12.19

JS)

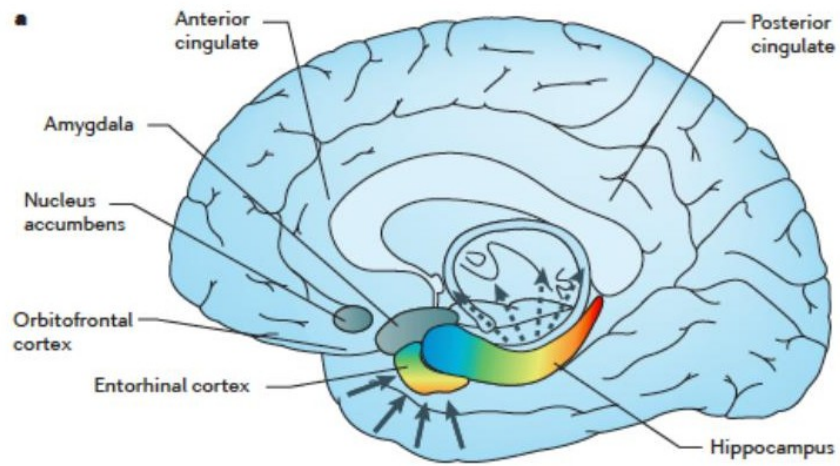


Figure L12.19 — Medial temporal regions affected in early AD